Appendix A v1.5: Positivity-Preserving Phase-Space Confinement, Compact Trapping Box Theorem, and Conditional Holographic Closure

Supplementary Artifact v1.5 (Upgraded Edition) — Additive to *Physics v1.0* and Appendices A v1.1–v1.4.1

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Abstract

Appendices A v1.1 through v1.4.1 established that the three-variable phenomenological dynamical system of *Physics v1.0* admits a unique interior equilibrium whose sole linear instability is a subcritical Hopf bifurcation ($l_1 > 0$) across the interior positive parameter orthant, and demonstrated that post-Hopf trajectories ($e > e_c$) escape the physical positive orthant $\mathbb{R}^3_+$ in finite time. This v1.5 supplement resolves that global phase-space breakdown by constructing a smooth (C^∞) soft-wall completion of the dynamical sector and proving a rigorous global trapping-box theorem. First, we formulate a conditional microphysical closure of Gate 0 via the covariant Hubeny–Rangamani–Takayanagi (HRT) and Bousso causal-diamond entropy bounds, explicitly separating conditional holographic identification from background-independent derivation. Second, by gating the irreversible-erasure sink with $\Psi_{\epsilon_H}(H_{\rm obs}) = \tanh(H_{\rm obs}/\epsilon_H)$ and the derivative-feedback channel with $\Phi_{\epsilon_I}(\dot{I}) = \tanh(\dot{I}/\epsilon_I)$, we prove via Nagumo's subtangentiality theorem that both the positive orthant $\mathbb{R}^3_+$ and the compact box $\mathcal{B} = [0, \gamma_{\max}] \times [0, H_{\max}] \times [0, \dot{I}_{\max}]$ are globally forward-invariant for all $t \geq 0$ and all positive parameters. Third, because $dH_{\rm obs}/dt|_* = 0$ at any steady state, the first derivative of the processing gate $\Phi_{\epsilon_I}(\dot{I})$ vanishes identically in the equilibrium Jacobian, leaving the interior equilibrium coordinates $(\gamma^*, \dot{I}^*)$ exact and preserving the v1.4 Jacobian, Hopf threshold e_c, and first Lyapunov coefficient $l_1 = +0.519589117351220\ldots$ up to exponentially small corrections from both gates at $\epsilon_H = \epsilon_I = 0.01$: $1 - \tanh(\dot{I}^*/\epsilon_I) = 2.77 \times 10^{-87}$ from the processing gate (yielding $\|J_{\rm v1.5} - J_{\rm v1.4}\|_F = 1.48 \times 10^{-86}$ and $|l_{1,\rm v1.5} - l_{1,\rm v1.4}| = 1.11 \times 10^{-82}$ via third-order gate derivatives) and $\mathrm{sech}^2(H^*_{\rm obs}/\epsilon_H) = 5.38 \times 10^{-695}$ from the entropy gate. All six boundary-invariance and ceiling-confinement conditions of $\mathcal{B}$ are formalized with complete machine-checked proofs in Lean 4 (`boundary_verification.lean`) and verified in arbitrary-precision arithmetic (`verification_patch_v1_5.py`).

1 Lineage Provenance and Epistemic Scope

Under the immutability rules of the *Physics v1.0* research lineage ([10.17605/OSF.IO/N3DQU]), the original unregularized ODE ansatz analyzed in Appendix A v1.1 and the exact symbolic positivity certificate of Appendix A v1.4/v1.4.1 (10.17605/OSF.IO/U7M6H, 10.17605/OSF.IO/S7WHX) remain frozen and unmodified. Specifically, Appendix A v1.1 identified two minimal options to address the finite-time escape to $\gamma < 0$ across the Hopf boundary $e = e_c$:

1. Restrict the admissible parameter domain to the linearly stable regime $e < e_c$ with the erasure admissibility bound $\dot{I}_{\rm erase} \leq c\gamma^* H_{\max}$ enforced; or
2. Introduce a versioned non-linear modification of the ansatz so that $\mathbb{R}^3_+ = \{(\gamma, H_{\rm obs}, \dot{I}) \in \mathbb{R}^3 \mid \gamma \geq 0,\ H_{\rm obs} \geq 0,\ \dot{I} \geq 0\}$ is globally forward-invariant while reproducing the v1.1–v1.4 equilibrium and Hopf scaling in the interior regime.

This document executes Option 2 as a separate versioned artifact (v1.5) and establishes a conditional holographic dictionary for the saturation capacity $H_{\max}$.

2 Conditional Holographic Dictionary for Gate 0

In *Physics v1.0*, the observer von Neumann entropy $H_{\rm obs}(t) = -\operatorname{Tr}(\rho_{\rm obs}\log_2\rho_{\rm obs})$ is bounded by an informational saturation capacity $H_{\max}$ (in bits). The Gate 0 Technical Report recorded the bridge between an abstract subsystem factorization $\mathcal{H}_{\rm total} = \mathcal{H}_{\rm obs}\otimes\mathcal{H}_{\rm env}$ and geometric horizon entropy $A/(4\ell_P^2\ln 2)$ as **UNRESOLVED IN GENERALITY** because an arbitrary algebraic tensor factorization need not correspond to a spatially localized region with a well-defined boundary area.

To make the microphysical assumption required to close Gate 0 mathematically explicit, we replace static Ryu–Takayanagi identification with a covariant causal-diamond localization postulate:

Proposition 1 (Conditional Holographic Closure of Gate 0)**.** *Let $(\mathcal{M}, g_{\mu\nu})$ be a $(3+1)$-dimensional semiclassical spacetime, and suppose the observer subsystem $\mathcal{H}_{obs}$ is defined by the algebra of quantum fields localized within the causal diamond $\mathcal{D}(\mathcal{A})$ of a spacelike ball $\mathcal{A}$ with smooth codimension-two boundary $\partial\mathcal{A}$. Under the covariant Bousso entropy bound and the Hubeny–Rangamani–Takayanagi (HRT) extremal surface prescription, if Σ_{HRT} is the unique minimal-area extremal surface homologous to $\mathcal{A}$ with $\partial\Sigma_{HRT} = \partial\mathcal{A}$ and stationary area* $\mathrm{Area}(\Sigma_{HRT}) = A_\Sigma$*, then the state-space supremum of the observer's fine-grained von Neumann entropy in bits is fixed by*

$$H_{\max} \equiv \sup_{\rho_{obs}\in\mathcal{S}(\mathcal{D}(\mathcal{A}))} [-\operatorname{Tr}(\rho_{obs}\log_2\rho_{obs})] = \frac{\mathrm{Area}(\Sigma_{HRT})}{4\ell_P^2\ln 2} \quad [bits]. \tag{1}$$

Remark 2 (Distinction Between Conditional Closure and General Derivation). Equation (1) closes Gate 0 *conditionally* within the class of spatially localized causal-diamond subsystems admitting a semiclassical holographic or covariant entropy bound. For abstract, non-geometric Hilbert-space factorizations $\mathcal{H}_{\rm total} = \mathcal{H}_{\rm obs}\otimes\mathcal{H}_{\rm env}$ without a spatial boundary $\partial\mathcal{A}$, Gate 0 remains open. None of the dynamical-sector theorems in Sections 3–6 depend on Eq. (1); they hold for any constant $H_{\max} > 0$.

3 Upgraded Soft-Wall Dynamical System and Global Trapping Box Theorem

In the unregularized v1.0 system, finite-time escape from $\mathbb{R}^3_+$ occurs through a two-stage boundary breach: (i) when $H_{\rm obs}$ approaches 0, the constant erasure term $-\dot{I}_{\rm erase}$ remains active and drives $H_{\rm obs} < 0$; and (ii) during large-amplitude oscillations above the subcritical Hopf threshold ($e > e_c$), negative entropy rates $dH_{\rm obs}/dt < 0$ enter the derivative-feedback term $e\,(dH_{\rm obs}/dt)$ unattenuated at $\dot{I} = 0$, driving $\dot{I} < 0$ and subsequently pulling $\gamma < 0$ via $a\dot{I}\mathcal{E}_{\rm avail}(1-\gamma/\gamma_{\max})$.

Note that replacing $-\dot{I}_{\rm erase}$ with $-\frac{1}{2}\dot{I}_{\rm erase}[1+\tanh(H_{\rm obs}/\epsilon_H)]$ or leaving $e\,(dH_{\rm obs}/dt)$ ungated at $\dot{I} = 0$ does *not* render $\mathbb{R}^3_+$ forward-invariant, because $\frac{1}{2}[1+\tanh(0)] = \frac{1}{2} \neq 0$ leaves a negative sink $-\frac{1}{2}\dot{I}_{\rm erase}$ at $H_{\rm obs} = 0$ whenever $\gamma < \dot{I}_{\rm erase}/(2cH_{\max})$, and an ungated $e\,(dH_{\rm obs}/dt)$ drives $\left.\frac{d\dot{I}}{dt}\right|_{\dot{I}=0} < 0$ whenever $e\,(dH_{\rm obs}/dt) < -f\mathcal{P}_{\rm input}$.

To achieve unconditional global forward-invariance of $\mathbb{R}^3_+$ via smooth (C^∞) real-analytic multipliers that vanish at $H_{\rm obs} = 0$ and $\dot{I} = 0$ and saturate to 1 in the interior, let $\epsilon_H, \epsilon_I > 0$ denote micro-scale boundary regularization widths ($\epsilon_H \ll H_{\max}$, $\epsilon_I \ll \dot{I}^*$) and define the gates

$$\Psi_{\epsilon_H}(H_{\rm obs}) \equiv \tanh\!\left(\frac{H_{\rm obs}}{\epsilon_H}\right), \qquad \Phi_{\epsilon_I}(\dot{I}) \equiv \tanh\!\left(\frac{\dot{I}}{\epsilon_I}\right). \tag{2}$$

The upgraded v1.5 dynamical system is governed by

$$\frac{d\gamma}{dt} = a\dot{I}\mathcal{E}_{\rm avail}\left(1-\frac{\gamma}{\gamma_{\max}}\right) - b\gamma, \tag{3}$$

$$\frac{dH_{\rm obs}}{dt} = c\gamma(H_{\max}-H_{\rm obs}) - \dot{I}_{\rm erase}\tanh\!\left(\frac{H_{\rm obs}}{\epsilon_H}\right), \tag{4}$$

$$\frac{d\dot{I}}{dt} = e\tanh\!\left(\frac{\dot{I}}{\epsilon_I}\right)\left(\frac{dH_{\rm obs}}{dt}\right) + f\mathcal{P}_{\rm input} - g\dot{I}, \tag{5}$$

where all parameters $a, b, c, e, f, g, \mathcal{E}_{\text{avail}}, \mathcal{P}_{\text{input}}, \dot{I}_{\text{erase}}, \gamma_{\max}, H_{\max}, \epsilon_H, \epsilon_I$ are strictly positive.

Theorem 3 (Global Forward-Invariance of $\mathbb{R}^3_+$ and Compact Trapping Box $\mathcal{B}$)**.** *For all strictly positive parameters and every feedback gain $e > 0$:*

1. *The non-negative orthant $\mathbb{R}^3_+ = \{(\gamma, H_{obs}, \dot{I}) \in \mathbb{R}^3 \mid \gamma \geq 0,\ H_{obs} \geq 0,\ \dot{I} \geq 0\}$ is positively invariant under the flow of Eqs. (3)–(5).*
2. *Define the maximum entropy-rate bound $M_H \equiv c\gamma_{\max} H_{\max} + \dot{I}_{erase}$ and the processing ceiling*

$$\dot{I}_{\max} \equiv \frac{eM_H + f\mathcal{P}_{input}}{g} > 0. \tag{6}$$

Then the compact box $\mathcal{B} \equiv [0, \gamma_{\max}] \times [0, H_{\max}] \times [0, \dot{I}_{\max}] \subset \mathbb{R}^3_+$ is positively invariant and globally absorbing within $[0, \gamma_{\max}] \times [0, H_{\max}] \times [0, \infty)$. Consequently, every trajectory originating in $\mathcal{B}$ exists globally for all $t \in [0, \infty)$ and never escapes to unphysical negative states or blows up in finite time.

Proof. Since the vector field $F = (d\gamma/dt, dH_{\text{obs}}/dt, d\dot{I}/dt)$ of Eqs. (3)–(5) is C^∞ (and therefore locally Lipschitz) on $\mathbb{R}^3$, by Nagumo's subtangentiality theorem a closed convex box $\mathcal{B} = \prod_{i=1}^3 [x_{i,\min}, x_{i,\max}]$ is positively invariant if and only if $F_i(x) \geq 0$ whenever $x \in \mathcal{B}$ with $x_i = x_{i,\min}$ and $F_i(x) \leq 0$ whenever $x \in \mathcal{B}$ with $x_i = x_{i,\max}$. We check all six boundary faces of $\mathcal{B}$ (and the three lower faces of $\mathbb{R}^3_+$):

1. **Face 1 — Lower γ-face ($\gamma = 0$, $\dot{I} \geq 0$):** Evaluating Eq. (3) at $\gamma = 0$ yields

$$\left.\frac{d\gamma}{dt}\right|_{\gamma=0} = a\dot{I}\mathcal{E}_{\text{avail}}(1 - 0) - b(0) = a\dot{I}\mathcal{E}_{\text{avail}} \geq 0, \tag{7}$$

   with strict positivity whenever $\dot{I} > 0$.

2. **Face 2 — Upper γ-ceiling ($\gamma = \gamma_{\max}$, $\dot{I} \geq 0$):** Evaluating Eq. (3) at $\gamma = \gamma_{\max}$ yields

$$\left.\frac{d\gamma}{dt}\right|_{\gamma=\gamma_{\max}} = a\dot{I}\mathcal{E}_{\text{avail}}\left(1 - \frac{\gamma_{\max}}{\gamma_{\max}}\right) - b\gamma_{\max} = -b\gamma_{\max} < 0. \tag{8}$$

3. **Face 3 — Lower H_{obs}-face ($H_{\text{obs}} = 0$, $\gamma \geq 0$):** Because $\tanh(0/\epsilon_H) = \tanh(0) = 0$, evaluating Eq. (4) at $H_{\text{obs}} = 0$ gives

$$\left.\frac{dH_{\text{obs}}}{dt}\right|_{H_{\text{obs}}=0} = c\gamma(H_{\max} - 0) - \dot{I}_{\text{erase}} \tanh(0) = c\gamma H_{\max} \geq 0 \quad \forall\gamma \geq 0. \tag{9}$$

   Notably, this non-negativity holds unconditionally for every $\dot{I}_{\text{erase}} > 0$ and every $\gamma \geq 0$, without requiring $\gamma(t) \geq \gamma^*$.

4. **Face 4 — Upper H_{obs}-ceiling ($H_{\text{obs}} = H_{\max}$, $\gamma \geq 0$):** Since $H_{\max}/\epsilon_H > 0 \implies \tanh(H_{\max}/\epsilon_H) > 0$,

$$\left.\frac{dH_{\text{obs}}}{dt}\right|_{H_{\text{obs}}=H_{\max}} = c\gamma(0) - \dot{I}_{\text{erase}} \tanh\left(\frac{H_{\max}}{\epsilon_H}\right) < 0. \tag{10}$$

5. **Face 5 — Lower $\dot{I}$-face ($\dot{I} = 0$, arbitrary $\gamma, H_{\text{obs}} \in \mathbb{R}$):** Because $\tanh(0/\epsilon_I) = 0$, both the feedback term $e\tanh(0)(dH_{\text{obs}}/dt)$ and the linear damping $-g(0)$ vanish identically at $\dot{I} = 0$:

$$\left.\frac{d\dot{I}}{dt}\right|_{\dot{I}=0} = e \cdot 0 \cdot \frac{dH_{\text{obs}}}{dt} + f\mathcal{P}_{\text{input}} - g(0) = f\mathcal{P}_{\text{input}} > 0. \tag{11}$$

6. **Face 6 — Upper $\dot{I}$-ceiling ($\dot{I} = \dot{I}_{\max}$ on $\mathcal{B}$):** For any $(\gamma, H_{\text{obs}}) \in [0, \gamma_{\max}] \times [0, H_{\max}]$, $dH_{\text{obs}}/dt \leq c\gamma_{\max} H_{\max} \leq c\gamma_{\max} H_{\max} + \dot{I}_{\text{erase}} = M_H$. Since $0 \leq \tanh(\dot{I}/\epsilon_I) \leq 1$, at $\dot{I} = \dot{I}_{\max}$ we have

$$\left.\frac{d\dot{I}}{dt}\right|_{\dot{I}=\dot{I}_{\max}} \leq eM_H + f\mathcal{P}_{\text{input}} - g\dot{I}_{\max} = 0. \tag{12}$$

Since F points inward on every boundary face of $\mathcal{B}$ and $\mathcal{B}$ is compact, standard ODE global existence guarantees that all trajectories in $\mathcal{B}$ are defined and confined to $\mathcal{B}$ for all $t \geq 0$. □

Key Structural Insight: In the post-Hopf unstable regime ($e > e_c$), the unique interior equilibrium loses stability via a subcritical Hopf bifurcation ($l_1 > 0$), while the boundary of the compact box $\mathcal{B} = [0, \gamma_{\max}] \times [0, H_{\max}] \times [0, \dot{I}_{\max}]$ is strictly inward-pointing. Instead of escaping to $-\infty$ in finite time as in v1.0, trajectories repelled from the equilibrium are trapped inside $\mathcal{B}$ and settle onto a globally bounded relaxation oscillation.

4 Formal Kernel Verification in Lean 4 (All Six Faces of $\mathcal{B}$)

To provide a complete machine-checked guarantee of Theorem 3, the accompanying Lean 4 file `boundary_verification.lean` formalizes Eqs. (3)–(6) over Mathlib's real numbers $\mathbb{R}$ and proves all six boundary face conditions (three lower orthant faces and three upper ceiling faces), an entropy-rate bound lemma (`dHobs_dt_le_M_H`), and the unified 6-face certificate (`compact_box_nagumo_certificate`) with zero `sorry` placeholders:

```
/-- Upper ceiling theorems (Faces 2, 4, 6) and lower face theorems (Faces 1, 3, 5). -/
theorem dHobs_dt_neg_at_ceiling
    (c H_max I_erase epsH : ℝ) (hIe : 0 < I_erase) (htanh : 0 < Real.tanh (H_max / epsH))
    (s : CosmologicalState) (hH : s.H_obs = H_max) :
    dHobs_dt c H_max I_erase epsH s < 0 := by
  unfold dHobs_dt
  rw [hH, sub_self, mul_zero, zero_sub]
  exact neg_lt_zero.mpr (mul_pos hIe htanh)

theorem dIdot_dt_nonpos_at_ceiling
    (e f g P_input c gamma_max H_max I_erase epsH epsI : ℝ)
    (he : 0 ≤ e) (hg : 0 < g) (hMH : 0 ≤ M_H c gamma_max H_max I_erase)
    (s : CosmologicalState)
    (hI : s.I_dot = I_dot_max e f g P_input c gamma_max H_max I_erase)
    (hdH : dHobs_dt c H_max I_erase epsH s ≤ M_H c gamma_max H_max I_erase)
    (htanh_nonneg : 0 ≤ Real.tanh (s.I_dot / epsI))
    (htanh_le_one : Real.tanh (s.I_dot / epsI) ≤ 1) :
    dIdot_dt e f g P_input c H_max I_erase epsH epsI s ≤ 0 := by
  unfold dIdot_dt I_dot_max at *
  have hg_ne : g ≠ 0 := ne_of_gt hg
  have h_g_mul : g * s.I_dot = e * M_H c gamma_max H_max I_erase + f * P_input := by
    rw [hI]
    exact mul_div_cancel0 (e * M_H c gamma_max H_max I_erase + f * P_input) hg_ne
  have h_step1 :
      Real.tanh (s.I_dot / epsI) * dHobs_dt c H_max I_erase epsH s ≤
      Real.tanh (s.I_dot / epsI) * M_H c gamma_max H_max I_erase :=
    mul_le_mul_of_nonneg_left hdH htanh_nonneg
  have h_step2 :
      Real.tanh (s.I_dot / epsI) * M_H c gamma_max H_max I_erase ≤
      1 * M_H c gamma_max H_max I_erase :=
    mul_le_mul_of_nonneg_right htanh_le_one hMH
  have h_step3 :
      Real.tanh (s.I_dot / epsI) * dHobs_dt c H_max I_erase epsH s ≤
      M_H c gamma_max H_max I_erase := by linarith
  have h_step4 :
      e * (Real.tanh (s.I_dot / epsI) * dHobs_dt c H_max I_erase epsH s) ≤
      e * M_H c gamma_max H_max I_erase :=
    mul_le_mul_of_nonneg_left h_step3 he
  have h_assoc :
      e * Real.tanh (s.I_dot / epsI) * dHobs_dt c H_max I_erase epsH s =
      e * (Real.tanh (s.I_dot / epsI) * dHobs_dt c H_max I_erase epsH s) :=
    mul_assoc e (Real.tanh (s.I_dot / epsI)) (dHobs_dt c H_max I_erase epsH s)
  linarith
```

5 Exact Equilibrium and Multi-Scale Spectral Invariance in the Interior

A critical advantage of gating only the erasure sink $\dot{I}_{\text{erase}} \tanh(H_{\text{obs}}/\epsilon_H)$ and the derivative-feedback term $e \tanh(\dot{I}/\epsilon_I)(dH_{\text{obs}}/dt)$—while keeping the linear relaxation terms $-b\gamma$ and $-g\dot{I}$ unperturbed—is that the interior steady-state coordinates $(\gamma^*, \dot{I}^*)$ are exact and the first-order gate derivative vanishes identically at equilibrium:

Proposition 4 (Exactness of $(\gamma^*, \dot{I}^*)$ and Multi-Scale Gate Perturbation Bounds)**.** *At any equilibrium $(\gamma^*, H^*_{obs}, \dot{I}^*)$ of Eqs. (3)–(5):*

1. *Since $dH_{obs}/dt|_* = 0$, Eq. (5) reduces at steady state to $f\mathcal{P}_{input} - g\dot{I}^* = 0$ and Eq. (3) reduces to $K(1-\gamma^*/\gamma_{\max}) - b\gamma^* = 0$ with $K \equiv a\mathcal{E}_{avail}\dot{I}^*$. Thus $\dot{I}^* = f\mathcal{P}_{input}/g$ and $\gamma^* = K\gamma_{\max}/(b\gamma_{\max} + K)$ are **identical** to Proposition 2 of Appendix A v1.1 with zero ϵ-dependence.*

2. *When linearizing Eq. (5) about the equilibrium, the product rule applied to $e \tanh(\dot{I}/\epsilon_I)(dH_{obs}/dt)$ yields*

$$\frac{\partial}{\partial \dot{I}}\left[e \tanh\left(\frac{\dot{I}}{\epsilon_I}\right)\frac{dH_{obs}}{dt}\right]\Bigg|_* = \frac{e}{\epsilon_I}\operatorname{sech}^2\left(\frac{\dot{I}^*}{\epsilon_I}\right)\underbrace{\frac{dH_{obs}}{dt}\Big|_*}_{=0} + e \tanh\left(\frac{\dot{I}^*}{\epsilon_I}\right)\underbrace{\frac{\partial(dH_{obs}/dt)}{\partial \dot{I}}\Big|_*}_{=0} = 0. \tag{13}$$

   Hence the $(3,3)$ entry of the Jacobian is exactly $-g$, with no $\operatorname{sech}^2(\dot{I}^*/\epsilon_I)$ linear damping artifact.

3. *In coordinates $(\gamma, H_{obs}, \dot{I})$, the exact Jacobian of the upgraded system (3)–(5) at equilibrium is*

$$J_{v1.5} = \begin{pmatrix} -\alpha & 0 & \beta \\ \delta_\epsilon & -\mu_\epsilon & 0 \\ e_{\mathit{eff}}\delta_\epsilon & -e_{\mathit{eff}}\mu_\epsilon & -g \end{pmatrix}, \tag{14}$$

   where α and β are identical to v1.1–v1.4, $e_{\mathit{eff}} = e \tanh(\dot{I}^*/\epsilon_I)$, $\delta_\epsilon = c(H_{\max} - H^*_{obs})$, and $\mu_\epsilon = c\gamma^* + \frac{\dot{I}_{erase}}{\epsilon_H}\operatorname{sech}^2(H^*_{obs}/\epsilon_H)$. Two distinct exponential gate scales govern the departure from v1.4:

   - ***Processing gate scale (ϵ_I):*** *$1 - \tanh(\dot{I}^*/\epsilon_I) \sim 2e^{-2\dot{I}^*/\epsilon_I}$, which dominates the Frobenius norm difference $\|J_{v1.5} - J_{v1.4}\|_F = O(e^{-2\dot{I}^*/\epsilon_I})$ (1.48×10^{-86} at $\dot{I}^* = 1, \epsilon_I = 0.01$) and, via the second and third derivatives of $\tanh(\dot{I}/\epsilon_I)$ in the normal-form bilinear/trilinear contractions, shifts l_1 by $O(\epsilon_I^{-2}\operatorname{sech}^2(\dot{I}^*/\epsilon_I)) = 1.11 \times 10^{-82}$.*
   - ***Entropy gate scale (ϵ_H):*** *$\operatorname{sech}^2(H^*_{obs}/\epsilon_H) \sim 4e^{-2H^*_{obs}/\epsilon_H}$, which shifts H^*_{obs} and μ_ϵ by $O(e^{-2H^*_{obs}/\epsilon_H})$ (5.38×10^{-695} at $H^*_{obs} = 8, \epsilon_H = 0.01$).*

6 High-Precision Verification and Global Post-Hopf Benchmark

All results are verified by `verification_patch_v1_5.py` using 60-digit and 750-digit `mpmath` arithmetic. Table 1 records the multi-scale spectral and Frobenius-norm verification at the v1.3/v1.4 audit slice ($a = 0.5, b = 1, c = 0.5, f = g = 1, \mathcal{E}_{\text{avail}} = 2, \mathcal{P}_{\text{input}} = 1, \dot{I}_{\text{erase}} = 0.5, \gamma_{\max} = 1, H_{\max} = 10$, with $\epsilon_H = \epsilon_I = 0.01$). Table 2 compares the global post-Hopf behavior of the upgraded v1.5 system against the finite-time escapes of Table 2 in Appendix A v1.1 ($a = b = c = f = g = \mathcal{E}_{\text{avail}} = 1, \gamma_{\max} = 10, H_{\max} = 100, \mathcal{P}_{\text{input}} = 5, \dot{I}_{\text{erase}} = 10, \epsilon_H = \epsilon_I = 0.05$).

Table 1: High-precision spectral, Frobenius-norm, and two-gate scale verification at the v1.3/v1.4 audit slice ($\epsilon_H = \epsilon_I = 0.01$).

Quantity	**Verified Value (60-Digit / 750-Digit `mpmath` Evaluation)**
Interior equilibrium $(\gamma^*, H^*_{\rm obs}, \dot{I}^*)$	$(0.5,\ 8.0,\ 1.0)$
Analytic Hopf threshold e_c	$135/26 = 5.192307692307692307692307692\ldots$
Hopf frequency ω_c (s^{-1})	$0.392232270276368063848324 93\ldots$
First Lyapunov coefficient $l_1 = \mathrm{Re}(G_{21})/(2\omega_c)$	$+0.519589117351220268983393 19\ldots$ (subcritical, $l_1 > 0$)
Eigenvector residual $\|Jq - i\omega_c q\|_\infty$	6.23×10^{-62}
Processing gate deviation $1 - \tanh(\dot{I}^*/\epsilon_I)$	$2.76779305347 \times 10^{-87}$ $(\dot{I}^*/\epsilon_I = 100)$
Entropy gate deviation $\mathrm{sech}^2(H^*_{\rm obs}/\epsilon_H)$	$5.38132158610 \times 10^{-695}$ $(H^*_{\rm obs}/\epsilon_H = 800)$
Soft-wall equilibrium shift $\lvert H^*_{\rm obs,v1.5} - H^*_{\rm obs,v1.4}\rvert$	$5.38132158610 \times 10^{-695}$ $(< 10^{-694} \ll 10^{-50})$
Jacobian Frobenius norm $\|J_{\rm v1.5} - J_{\rm v1.4}\|_F$	$1.48135280746 \times 10^{-86}$ $(< 10^{-85} \ll 10^{-50})$
Hopf threshold shift $\lvert e_{c,\rm v1.5} - e_{c,\rm v1.4}\rvert$	$1.43712331623 \times 10^{-86}$ $(< 10^{-85} \ll 10^{-50})$
First Lyapunov shift $\lvert l_{1,\rm v1.5} - l_{1,\rm v1.4}\rvert$	$1.10793084174 \times 10^{-82}$ $(< 10^{-81} \ll 10^{-50})$
Linear gate derivative $\partial_{\dot{I}}[e\tanh(\dot{I}/\epsilon_I)(dH_{\rm obs}/dt)]_*$	0 (vanishes identically since $dH_{\rm obs}/dt\|_* = 0$)

Table 2: Global post-Hopf trajectory confinement on the v1.1 benchmark ($\dot{I}_{\rm erase} = 10, e_c = 4.488095, \epsilon_H = \epsilon_I = 0.05$). Every trajectory that escaped to $\gamma < 0$ in finite time under v1.0 is confined strictly inside $\mathcal{B} = [0, \gamma_{\max}] \times [0, H_{\max}] \times [0, \dot{I}_{\max}]$ under v1.5.

Gain	**v1.0 / v1.1 Behaviour**	**v1.5** $\min\gamma(t)$	**v1.5** $\max\gamma(t)$	**v1.5** $\min H_{\rm obs}(t)$	**v1.5 Global Regime** ($t \in [0, 400]$)
$e = 0.90\, e_c$	Decays to eq. (6)	3.333272	3.333433	96.999947	Asymptotically stable equilibrium
$e = 1.01\, e_c$	Bounded osc. (5.4×10^{-3})	3.331644	3.334981	96.998534	Bounded interior oscillation
$e = 1.04\, e_c$	Escapes: $\gamma \to 0^-$ at $t \approx 259$	0.005682	9.720486	0.003339	Globally confined relaxation cycle in $\mathcal{B}$
$e = 1.08\, e_c$	Escapes: $\gamma \to 0^-$ at $t \approx 136$	0.005461	9.730970	0.003222	Globally confined relaxation cycle in $\mathcal{B}$
$e = 1.20\, e_c$	Escapes: $\gamma \to 0^-$ at $t \approx 55$	0.004889	9.758112	0.002921	Globally confined relaxation cycle in $\mathcal{B}$
$e = 2.00\, e_c$	Escapes: $\gamma \to 0^-$ at $t \approx 19$	0.002885	9.854966	0.001854	Globally confined relaxation cycle in $\mathcal{B}$

References

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